

Software Requirements Specification (SRS)

Project: Hospital Management System (HMS) – Patient Record & Appointment Management

Team: 13

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Hospital Management System (HMS) being developed by Team 13. It defines the functional and non-functional requirements, interfaces, and verification criteria for a patient record and appointment management system for hospitals, and serves as the baseline for design, implementation, and testing.

1.2 Scope

The HMS covers front-desk registration of patients, maintenance of patient medical records, scheduling and management of doctor appointments, doctor/staff record management, prescription and consultation logging, and basic billing for consultations. The system is implemented in C/C++ as a standalone or LAN client-server desktop application backed by a local relational data store (SQLite). It excludes full hospital-wide ERP functions such as pharmacy inventory, insurance claim processing, and radiology/lab information systems, which are out of scope for this mini-project.

1.3 Audience

Developers (Team 13), QA reviewers, the course instructor/evaluator, and future maintainers of the codebase.

1.4 Definitions

HMS – Hospital Management System; OPD – Out-Patient Department; UI – User Interface; CLI – Command-Line Interface; DB – Database; RBAC – Role-Based Access Control; RTM – Requirements Traceability Matrix; FR – Functional Requirement; NFR – Non-Functional Requirement.

2. Overall Description

2.1 Product Perspective

HMS is a self-contained desktop/LAN application used at hospital front-desk and doctor terminals. It is composed of a C/C++ application layer, a data access layer, and an embedded SQLite database. In its multi-terminal configuration, a lightweight TCP socket server (written in C) allows several front-desk/doctor clients on the same hospital LAN to operate against a single shared database.

2.2 Major Product Functions (detailed)

- Staff authentication and role-based access (Admin, Doctor, Receptionist)
- Patient registration and record management
- Appointment booking, rescheduling, and cancellation
- Doctor availability and schedule management
- Consultation, diagnosis, and prescription logging
- Billing for consultations and payment status tracking
- Daily/weekly reporting and database backup
- System administration and health checks

2.3 User Roles and Characteristics (expanded)

- Receptionist: registers patients, books/cancels appointments, generates bills; minimal technical background, needs a simple menu-driven interface.
- Doctor: views own schedule/queue, records diagnosis and prescriptions; expects fast lookups during consultations.
- Administrator: manages staff accounts, doctor records, views reports, performs backups and system health checks.

2.4 Operating Environment

Windows or Linux desktop/workstation, standard hospital front-desk hardware, optional LAN connectivity between terminals, local power backup (UPS) recommended for front-desk machines.

2.5 Constraints

Implementation restricted to C/C++ (per course requirement). Patient data must be handled with confidentiality safeguards appropriate for medical records. No dependency on external cloud services; the system must run fully on hospital-owned infrastructure.

3. External Interface Requirements

3.1 User Interfaces

Primary UI: menu-driven console interface (with an optional Qt/GTK GUI as a stretch goal) exposing role-specific menus. Clear prompts and confirmation messages for destructive actions (e.g., cancelling an appointment, deleting a record).

3.2 Hardware Interfaces

- Standard desktop/laptop workstation (keyboard, monitor)
- Optional receipt/bill printer at the front desk
- Optional barcode/ID-card reader for patient lookup (future enhancement)

3.3 Software Interfaces

- SQLite embedded database engine (via the C SQLite3 API) for persistent storage
- POSIX/Winsock TCP sockets for the optional multi-terminal client-server mode
- Local file system for backups, logs, and exported reports (CSV)

3.4 Communications

In standalone mode, all communication is in-process function calls to the data access layer. In LAN mode, clients communicate with the HMS server over TCP using a lightweight length-prefixed request/response protocol; all client sessions require successful authentication before any data request is served.

[Note: overall this SRS defines 22 Functional Requirements, 6 Non-Functional Requirements, 2 Security Objectives, and 5 Security Requirements, consistent with the minimum required by the assignment (15 FRs, 5 NFRs, 2 security objectives, 5 security requirements).]

4. System Features (detailed)

Each requirement below includes acceptance criteria and a reference test case ID. Requirement IDs follow HMS-F-####.

4.1 Authentication & Access Control

Description: Authenticate staff and enforce role-based access to system functions.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Ref	Comments/Dependencies
HMS-F-001	The system shall authenticate staff via a unique username and password before granting access to any function.	Functional	High	Security	AC-001: Valid credentials → role-specific menu shown. Test: TC-Auth-01	Requires user account table
HMS-F-002	The system shall enforce role-based access control (Admin, Doctor, Receptionist) restricting	Functional	High	Admin/Security	AC-002: Receptionist cannot access doctor-only prescription entry. Test: TC-Auth-02	Depends on HMS-F-001

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Ref	Comments/Dependencies
	each menu action to permitted roles.					
HMS-F-003	The system shall lock a staff account for 5 minutes after 3 consecutive failed login attempts and record an audit event.	Functional	High	Security	AC-003: 4th attempt within lockout is rejected with a lockout message. Test: TC-Auth-03	Audit log required

4.2 Patient Record Management

Description: Create, retrieve, update, and archive patient demographic and medical-history records.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Ref	Comments/Dependencies
HMS-F-010	The system shall allow a receptionist to register a new patient with name, age, gender, contact number, address, and blood group.	Functional	High	Front Desk	AC-010: New patient assigned a unique Patient ID. Test: TC-Pat-01	Patient ID auto-generated
HMS-F-011	The system shall allow searching for a patient record by Patient ID, name, or phone number.	Functional	High	Front Desk/Doctor	AC-011: Search returns matching record(s) within 2s. Test: TC-Pat-02	Depends on HMS-F-010
HMS-F-012	The system shall allow authorized staff to update an existing patient's contact and demographic details.	Functional	Medium	Front Desk	AC-012: Updated fields persist after re-query. Test: TC-Pat-03	
HMS-F-013	The system shall allow a doctor to view a patient's full visit and prescription history.	Functional	High	Doctor	AC-013: History listed in reverse chronological order. Test: TC-Pat-04	Depends on HMS-F-016
HMS-F-014	The system shall allow an Admin to soft-delete (archive) a patient record instead of permanent deletion.	Functional	Medium	Admin	AC-014: Archived record excluded from active search but retrievable by Admin. Test: TC-Pat-05	Admin-only

4.3 Appointment Management

Description: Book, view, reschedule, and cancel appointments while checking doctor availability.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Ref	Comments/Dependencies
HMS-F-020	The system shall allow booking an appointment for a patient with a chosen doctor, date, and time slot.	Functional	High	Front Desk	AC-020: Appointment created only if the slot is free. Test: TC-Appt-01	Depends on HMS-F-030
HMS-F-021	The system shall check doctor availability before confirming a booking and	Functional	High	Front Desk	AC-021: Double-booking attempt is rejected with a message. Test: TC-Appt-02	Depends on HMS-F-020

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Ref	Comments/Dependencies
	reject conflicting bookings.					
HMS-F-022	The system shall allow a receptionist to reschedule or cancel an existing appointment.	Functional	High	Front Desk	AC-022: Status updated to Rescheduled/Cancelled; slot freed. Test: TC-Appt-03	
HMS-F-023	The system shall display a doctor's daily/weekly appointment schedule sorted by time.	Functional	High	Doctor/Front Desk	AC-023: Schedule list matches DB records for selected date range. Test: TC-Appt-04	
HMS-F-024	The system shall print/display an appointment confirmation slip with patient, doctor, date, and time.	Functional	Medium	Front Desk	AC-024: Slip generated immediately after booking. Test: TC-Appt-05	Depends on HMS-F-020

4.4 Doctor & Staff Management

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Ref	Comments/Dependencies
HMS-F-030	The system shall allow an Admin to add, update, or remove doctor profiles including specialization and working hours.	Functional	High	Admin	AC-030: New doctor appears in booking doctor list. Test: TC-Doc-01	
HMS-F-031	The system shall allow an Admin to create staff (Receptionist/Doctor) accounts with an assigned role.	Functional	High	Admin	AC-031: New account can log in with assigned role only. Test: TC-Doc-02	Depends on HMS-F-001
HMS-F-032	The system shall display each doctor's current patient queue/workload for the day.	Functional	Medium	Doctor	AC-032: Queue count matches booked appointments for today. Test: TC-Doc-03	

4.5 Consultation & Prescription

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Ref	Comments/Dependencies
HMS-F-040	The system shall allow a doctor to record diagnosis notes and a prescription for a patient's visit.	Functional	High	Doctor	AC-040: Prescription saved and linked to the visit and patient. Test: TC-Rx-01	Depends on HMS-F-020
HMS-F-041	The system shall allow retrieval of a patient's prescription history by visit date.	Functional	Medium	Doctor/Front Desk	AC-041: All past prescriptions listed chronologically. Test: TC-Rx-02	Depends on HMS-F-040

4.6 Billing

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Ref	Comments/Dependencies
HMS-F-050	The system shall generate a consultation bill based on doctor consultation fee after a visit is completed.	Functional	High	Front Desk	AC-050: Bill total matches configured fee for the doctor. Test: TC-Bill-01	Depends on HMS-F-040
HMS-F-051	The system shall record and update payment status (Paid/Pending) for each bill.	Functional	Medium	Front Desk	AC-051: Status change persists and reflects in reports. Test: TC-Bill-02	Depends on HMS-F-050

4.7 Reporting & Administration

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Ref	Comments/Dependencies
HMS-F-060	The system shall generate a daily appointment report listing all appointments and their status.	Functional	Medium	Admin	AC-060: Report count matches DB for the selected date. Test: TC-Rep-01	
HMS-F-061	The system shall allow an Admin to export the patient database to a backup file on demand.	Functional	High	Admin	AC-061: Backup file created and restorable. Test: TC-Rep-02	File I/O
HMS-F-062	The system shall provide an Admin-only health check reporting database connectivity and available disk space.	Functional	Medium	Admin	AC-062: Health check returns OK/FAIL status accurately. Test: TC-Rep-03	

5. Non-Functional Requirements (detailed)

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Ref	Comments/Dependencies
HMS-NF-001	The system shall complete any single patient/appointment lookup within 2 seconds for a database of up to 10,000 patient records.	Performance	High	All Users	AC-NF-001: Measured lookup time < 2s under test load. Test: TC-Perf-01	
HMS-NF-002	The system shall be available during hospital operating hours with no more than 1 unplanned restart per week.	Availability	High	Hospital Ops	AC-NF-002: Uptime log shows compliance over a 1-week test window. Test: TC-Avail-01	
HMS-NF-003	The system's menus and prompts shall be usable by non-technical staff with less than 30 minutes of training.	Usability	Medium	Front Desk	AC-NF-003: Usability walkthrough completed without external help. Test: TC-Use-01	
HMS-NF-004	The system shall not lose committed data on an abrupt process termination or power loss (transactional writes).	Reliability	High	All Users	AC-NF-004: Data intact after simulated crash during a write. Test: TC-Rel-01	SQLite transactions

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Ref	Comments/Dependencies
HMS-NF-005	The system shall build and run on both Windows and Linux without code changes (portable C/C++).	Portability	Medium	Developers	AC-NF-005: Successful build/run on both platforms. Test: TC-Port-01	
HMS-NF-006	The codebase shall be organized into clearly separated modules (UI, business logic, data access) to support maintainability.	Maintainability	Medium	Developers	AC-NF-006: Module boundaries verified in code review. Test: TC-Maint-01	

Security Objectives

- SO-1: Protect the confidentiality of patient medical and personal data from unauthorized access.
- SO-2: Ensure accountability by making all access to and modification of patient/appointment data attributable to an authenticated staff member.

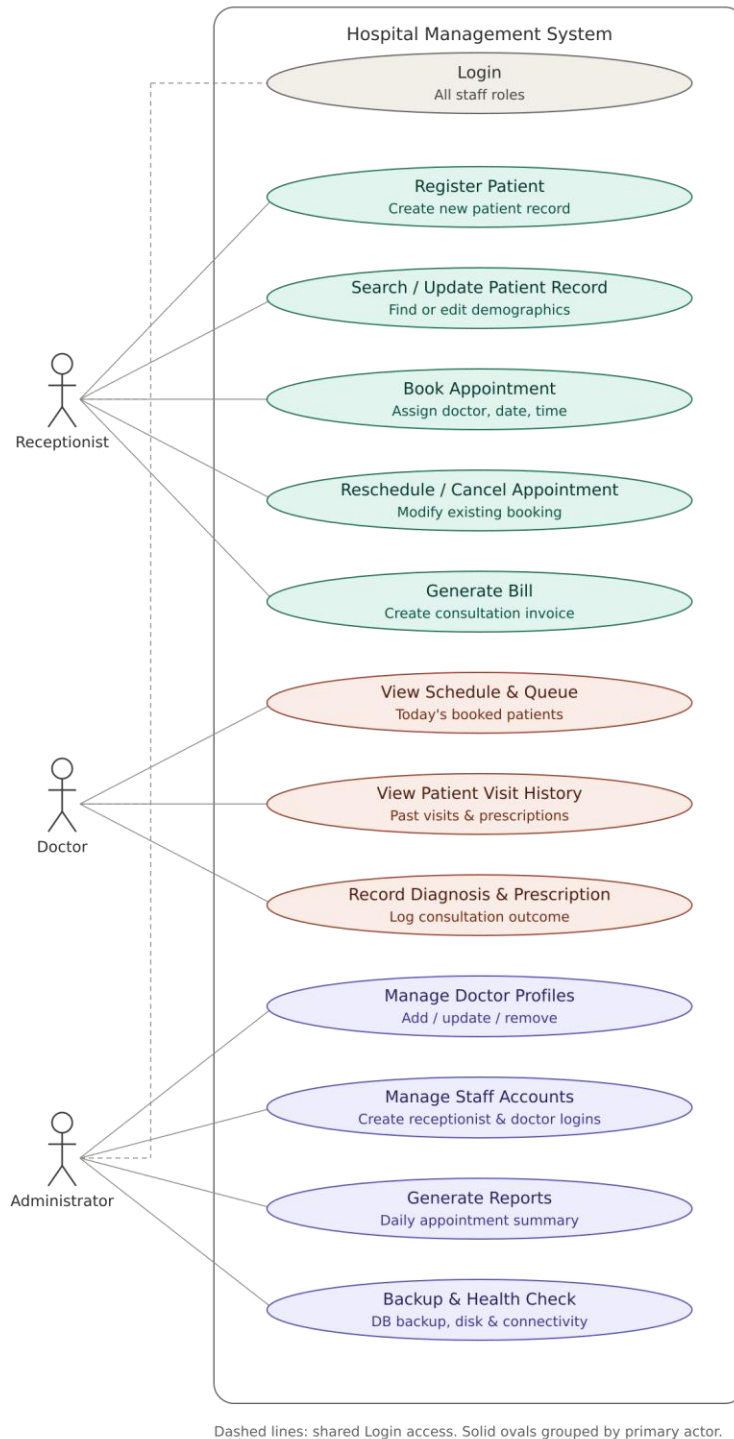
Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Ref	Comments/Dependencies
HMS-SR-001	The system shall store staff passwords as salted cryptographic hashes and never in plaintext.	Security	High	Security	AC-SR-001: DB inspection shows no plaintext passwords. Test: TC-Sec-01	
HMS-SR-002	The system shall enforce role-based access control at every data-access function, not only at the UI layer.	Security	High	Security	AC-SR-002: Direct function call with wrong role is rejected. Test: TC-Sec-02	Depends on HMS-F-002
HMS-SR-003	The system shall log every create/update/delete on patient, appointment, and prescription records with timestamp and user ID.	Security	High	Security/Compliance	AC-SR-003: Audit log contains an entry for each tested action. Test: TC-Sec-03	
HMS-SR-004	The system shall validate and bound-check all user input (avoiding unsafe C functions such as gets/strcpy) to prevent buffer overflows and injection into the data layer.	Security	High	Security	AC-SR-004: Static analysis/fuzz test of input fields reports no overflow. Test: TC-Sec-04	Use of parameterized SQL
HMS-SR-005	The system shall automatically log out a staff session after 10 minutes of inactivity.	Security	Medium	Security	AC-SR-005: Session terminated after idle timeout in test. Test: TC-Sec-05	

6. Quality Attributes & Acceptance Tests

Acceptance for this release requires: all High-priority functional requirements implemented and passing their referenced test cases; zero critical defects open; all 5 security requirements verified; NFR-001 and NFR-004 demonstrated under test; and successful build on both target platforms (NFR-005).

7. UML Use-Case Diagram



Primary actors: Receptionist, Doctor, Administrator. Key use cases: Login (shared by all roles), Register Patient, Search/Update Patient Record, Book Appointment, Reschedule/Cancel Appointment, Generate Bill, View Schedule & Queue, View Patient Visit History, Record Diagnosis & Prescription, Manage Doctor Profiles, Manage Staff Accounts, Generate Reports, and Backup & System Health Check.

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement Summary	Design Component	Test Case ID
HMS-F-001	Staff authentication	Auth Module	TC-Auth-01
HMS-F-010	Patient registration	Patient Manager	TC-Pat-01
HMS-F-020	Book appointment	Appointment Manager	TC-Appt-01
HMS-F-030	Manage doctor profiles	Doctor Manager	TC-Doc-01
HMS-F-040	Record prescription	Consultation Manager	TC-Rx-01
HMS-F-050	Generate bill	Billing Manager	TC-Bill-01
HMS-SR-001	Password hashing	Auth Module	TC-Sec-01
HMS-SR-003	Audit logging	Audit Logger	TC-Sec-03

(Full RTM to be expanded to cover all requirement IDs as design and test cases are finalized.)